ML Challenge 2026: Business Entity Resolution Solution Template

Team Name: [Your Team Name]
Team Members: [List all team members]
Submission Date: 2026-09-26

---

1. Executive Summary

We resolve Source-2/Source-3 records to Source-1 entities with a classic *blocking → pairwise classifier → graph-aware decision* pipeline, engineered around the macro-F0.5 metric. Blocking uses pruned TF-IDF nearest-neighbour search on unigram+bigram tokens of name and
address in **both directions** (each S2/S3 record retrieves its likely S1 parents, each S1 retrieves its likely children), reaching a 99.5 % recall ceiling with ~60 candidates per S1 entity. A two-stage LightGBM model scores pairs: stage 1 from ~70 string/structure similarity features, stage 2 adds *contextual* features describing how the pair ranks among the competing candidates of the same S1 entity and of the same S2/S3 record (every S2/S3 record has at most one parent). Final match lists are chosen per entity by maximising the *expected* F0.5 under the model's calibrated probabilities, which naturally applies a stricter bar to each additional match and to singletons. Validation macro-F0.5 on a held-out, test-like split is **0.9901**.

---

2. Methodology

2.1 Problem Analysis

Findings from exploratory analysis of the training data (2.2 M S1, 5.0 M S2, 5.3 M S3 records):

- **Structure.** Every S2/S3 record is matched to at most one S1 entity (7.64 M matches, no ID appears twice in the ground truth). Only 5.6 % of S1 entities are singletons; the mean entity has 3.5 matches (max 11). ~26 % of S2/S3 records match nothing ("distractors"). No match crosses a country boundary.
- **Distractors are hard negatives.** Unmatched S2/S3 records are near-clones of S1 entities: same name with a different legal suffix and a *slightly different house number* (7686 vs 7681, 378 vs 357), or the same name in a different city. Name similarity alone is therefore

useless for precision — the median distractor is *more* name-similar to its nearest S1 than the median true match. Address structure (house number, street, locality) carries the discriminative signal.

- **Names are highly ambiguous.** 36 % of US and 44 % of India S1 records share their exact normalised name with another S1 record (e.g. 253 × "Primary Care Group").
- **Blank addresses** (3.4 % of S2/S3) are almost always true matches (97.8 %) — the generator blanks addresses of genuine matches — but their parent must be identified from the name alone.
- **Noise patterns.** Names: case, typos (incl. digit-for-letter "Nati0nal"), diacritics, legal-suffix add/drop/reorder, word shuffles, junk prefixes ("--", ">>", "M/s", "Shri"), junk suffixes ("Services", "#87529", "(ID: 51621)"), "formerly"/"D.B.A."/"née" renames, domain-name forms ("telecomnirman.com"), duplicated words, and full transliteration of Indian names into nine Indic scripts (14 % of India S2 names). Addresses: abbreviations, component reordering, missing components, leading zeros / suffixes on house numbers ("0050", "132-C", "627 1/2", "24035-24039"), city ↔ neighbourhood substitutions (Sacramento ↔ Foothill Farms), state as full name / abbreviation / native script.
- **Test set** adds France (Latin script with diacritics, French legal forms SARL/SAS/SASU/SCI, region/department inconsistencies) which never appears in training.

2.2 Solution Strategy

Approach Type: Blocking + two-stage pairwise classifier + expected-F0.5 decision
Core Innovation: (i) bidirectional pruned TF-IDF blocking with name/address bigrams; (ii) stacked *contextual* features that let the model see the competition for each S2/S3 record (one-parent structure) and for each S1 entity; (iii) a decision rule that maximises expected macro-F0.5 per entity instead of thresholding.

Everything is language/country agnostic: no hard-coded suffix lists or country rules. The only learned lexical resource is a token dictionary (Indic-script token → Latin token) learned by co-occurrence from matched training pairs; unknown tokens fall back to `unidecode`. "Generic" name tokens (inc, llc, private, limited, sarl, …) are derived unsupervised from each split's own document frequencies, so France is handled without special casing.

---

3. Candidate Generation (Blocking)

Per country (no cross-country matches exist), four TF-IDF nearest-neighbour passes (`sparse_dot_topn`, cosine, float32, 10 threads):

pass	index	direction	top-k
R	name+address unigrams & bigrams, `max_df=0.05`	each S2/S3 → S1	10
F	same index	each S1 → S2/S3	20
A	address unigrams & bigrams	each S2/S3 with an address → S1	5
N	name char-3-grams	each S2/S3 with a *blank* address → S1	20

- **Blocking keys used:** TF-IDF-weighted unigrams and bigrams of the normalised name and address (leading zeros stripped, digit/letter boundaries split, foreign tokens mapped through the learned dictionary). Bigrams are the key trick: in India, unigrams such as *first, private, mumbai, 8* all exceed the document-frequency cutoff, but *first_food* survives.
- **Candidate pairs generated:** dev split 34.3 M pairs for 551 k S1 entities (62 per S1); test 126.7 M (France 19.5 M, India 56.2 M, US 51.0 M; 73 per S1) pairs for 1.73 M S1 entities.
- **How we ensured true matches were not lost:** recall of the union is measured on the dev split against ground truth: **99.50 %** (US 99.65 %, India 99.26 %). Ablations: R alone 97.3 %, F alone 96.4 %, A alone 88.9 %; the reverse direction is essential because for common names the S1→S2/S3 top-k is exhausted by near-identical clones of *other* entities, whereas each S2/S3 record's own top-10 almost always contains its parent (top-1 = 96 %). Remaining misses are initials-only domain names ("gwk.com" for "WK Garden Limited"), synthetic replacement names with a garbled address, and blank-address records whose typo'd
  name collides with dozens of near-identical S1 names.

Cost: dev ≈ 15 min, test ≈ 4 h wall-clock on the 10-core laptop (India alone 3.4 h because its 4.7 M records × 0.8 M S1 columns make the top-k products expensive; the run overlapped with model training). The union of all four passes is exactly the set
written to `candidate_pairs.tsv` and scored by the model.

---

4. Matching Model

Features used (74, all generic):

- Name features: TF-IDF cosine (word, char-3-gram), RapidFuzz ratio / token-sort / token-set / partial ratio, Jaro-Winkler, normalised Levenshtein, token Jaccard and overlap

counts, the same on the *core* name (generic high-DF tokens removed), first-token equality, subset relation, missing/extra core tokens, typo-tolerant token coverage, no-space partial ratio (catches domain-name forms), foreign-script fraction, domain flag, and the number of S1 entities in the country sharing the exact name / core name (ambiguity).

- Address features: TF-IDF cosine on address words, token-set/sort/partial/ratio, alpha-token Jaccard and overlap, last-token equality/containment (state/city), typo-tolerant token coverage both ways, blank flag, foreign fraction; **house-number features**: equality, Levenshtein distance between the digit strings, length difference, prefix relation, absolute numeric difference, presence of one number in the other address, numeric-token Jaccard and
  conflict flags; numeric tokens in names (branch ids) conflict/extra.
- Other: source flag (S2/S3), blocking-pass indicators, combined name+address cosine.
- Contextual (stage 2, from stage-1 scores): for the S1 entity — max score, rank, gap to max, margin to second best, counts of candidates above 0.2/0.5, candidate count, same-source max
  and count; for the S2/S3 record — max score over all its S1 candidates, rank, gap, margin to
  the second-best S1, counts above 0.2/0.5. `ctx_s23_margin` and `ctx_s23_rank` are the two
  most important stage-2 features: they encode the one-parent constraint softly.

Model type: two stacked LightGBM binary classifiers (num_leaves 127, lr 0.05, early stopping on an S1-grouped holdout). Training pairs come from a dev split that mirrors the test structure (25 % of S1 entities, all their matches, 25 % of distractors). All positives and all *hard* negatives (plausible cosine overlap or blank address) are kept; easy negatives are sampled at 10 % with inverse-probability weights, so probabilities stay calibrated (reliability check: predicted 0.7–0.8 → 76 % true). Stage-2 training data uses out-of-fold stage-1 scores (3 folds by S1 entity) to avoid leakage.

Threshold selection method: none in the usual sense. For every S1 entity the candidates are sorted by probability and, after the one-parent filter (an S2/S3 record is kept only for its highest-probability S1), the prefix that maximises the *expected* F0.5 under independent Bernoulli outcomes is selected (exact Poisson-binomial computation). This adds a first match at $p \gtrsim 0.5$, a second only at $p \gtrsim 0.73$, a fourth at $p \gtrsim 0.77$, and keeps an entity a singleton unless a candidate is convincing — exactly the asymmetry F0.5 rewards. It scored 0.9901 vs 0.9899 (threshold 0.7) for the best fixed threshold on validation.

---

5. Results & Error Analysis

- **F0.5 Score (macro):** 0.9901 on 275 857 held-out dev S1 entities (US 0.9907, India 0.9892); singletons score 0.9892. Stage 1 alone: 0.9864.
- **Common false positives (wrong merges):** clone distractors whose only difference from the entity is a small house-number change (208/1 vs 208/2, 10201 vs 0203) or a one-word suffix swap; a few blank-address records whose name equals a *different* S1 entity with the same name (irreducible without an address). False positives account for only ~27 % of the lost score.
- **Common false negatives (missed matches):** ~73 % of the loss. 61 % of missed-in-candidate matches are blank-address records whose truncated / typo'd name has several plausible S1 parents (the model correctly abstains when the margin is small); records renamed to a synthetic word or an initials-only domain; ~19 % of misses were never generated as candidates.
- Probability calibration on validation is close to the diagonal, which is what makes the expected-F0.5 decision work.
- **Generalisation to the unseen country (France, test set):** without any France-specific code the pipeline produces the same match statistics as in training — 95.3 % of S1 entities receive at least one match (train 94.4 %), 3.50 matches per entity (train 3.46), and the matched records include the same noise families (renames "Zetadeltaevo trading as Breizh Soins SARL", domain forms "closlycesas.com", typos "R FULOTN", region ↔ department swaps Nouvelle-Aquitaine ↔ Gironde). Near-clone distractors (same name, house number 5 vs 1, or a different street) receive p ≤ 0.4 and are left unmatched.

Test-set output (`output/matching_results.tsv`): 1 732 544 S1 rows, 94 265 left empty (5.4 % singletons vs 5.6 % in training), 5 905 293 matches in total (3.41 per entity vs 3.46 in training); `candidate_pairs.tsv` holds the 126.7 M scored pairs. `utils/validate_submission.py --check-ids` reports PASS.

6. Conclusion

A carefully engineered classical pipeline — bidirectional bigram TF-IDF blocking, a rich but generic feature set centred on address structure, stacked contextual features that exploit the one-parent constraint, and an expected-F0.5 decision rule — resolves millions of noisy records on a laptop with a validation macro-F0.5 of 0.9901. The main lessons: precision on this data comes from house numbers and localities rather than names, the reverse retrieval

direction is what makes blocking robust to ambiguous names, and calibrated probabilities plus a metric-aware decision beat any fixed threshold.

Appendix

A. Code Artefacts

`code/business_entity_resolution/` contains `src/ber/` (package), `README.md` with the exact
command sequence, and `requirements.txt` (pinned). Entry points, in order:
`make_dev → learn_translit → prep → blocking → label → features → train_eval` (dev) and
`prep → blocking → predict → write_output` (test). `predict.py` reproduces `output/candidate_pairs.tsv` (the exact scored pair set) and `output/matching_results.tsv`.

B. Additional Results

Blocking ablation on the dev split (recall of true pairs among candidates):

configuration	**recall**	**pairs / S1**
name-only char-3-gram top-20 (S1→S2/S3)	55 %	20
name+address unigrams `max_df=0.01`, 4 passes	98.2 %	64
+ name/address bigrams, `max_df=0.05`	**99.5 %**	62

Validation macro-F0.5 by decision rule:

decision rule	**macro-F0.5**
fixed threshold 0.5	0.9885
fixed threshold 0.7	0.9899
fixed threshold 0.8	0.9898
expected-F0.5 prefix (no one-parent filter)	0.9901
expected-F0.5 prefix + one-parent filter	**0.9901**
stage 1 only, expected-F0.5	0.9864

Stage 1 with uniform 20 % negative sampling scored 0.9858; keeping every *hard* negative (the hard-negative sampling described in §4) raised it to 0.9864 and the stacked model from 0.9893 to 0.9901.